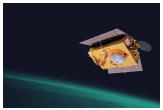
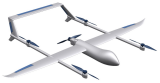
	Platform	Illustration	Ref	Advantages	Disadvantages
Stationary	Structure		[68]	Continuous and real-time monitoring High resolution using standard sensors Stable data collection Low cost for small-scale areas	Limited coverage area Maintenance requirements Requires installation Expensive to expand coverage
	Satellite		[69]	Global coverage Minimal maintenance Cost-effective for large areas On-demand service	Low resolution Lag in data acquisition High initial cost Dependency on weather conditions
Manned	Foot Patrol		[70]	Adaptability to diverse environments Cost-effective for dense asset areas Flexibility on the data detail Access to hard-to-reach areas	Limited coverage for widespread assets Safety risks at heights or live assets Labor-intensive for large-scale inspections Slow inspection speed over long distances
	Road Vehicle		[71]	Faster data collection for distributed assets Long-range capability and wide-area coverage Supports heavy equipment and multiple sensors Enables frequent and passive data acquisition	Limited to assets near drivable paths Cannot inspect elevated structures effectively Lower flexibility in navigating obstacles Slower than aerial methods
	Helicopter		[72]	Rapid inspection over large distances Access to remote and difficult terrain High-resolution imaging from aerial perspective High payload capacity of sensing equipment	High operational and maintenance costs Weather-dependent flight conditions Requires specialized pilots and permissions Limited low-altitude stability for precise data
Unmanned	Climbing Robot		[73]	Access to live power lines Enables detailed conductor inspections Reduces human exposure to high-risk areas Can operate autonomously in certain cases	Limited to power lines or similar structures Complex deployment and retrieval Slow and limited mobility Potential wear on conductors over time
	Multi-Rotor		[74]	High maneuverability in tight spaces Provides detailed aerial imaging Access to hard-to-reach areas Rapid deployment with minimal setup	Limited flight time and range Weather-sensitive performance Requires skilled operator Low payload capacity
	Fixed-Wing		[75]	Efficient for large area coverage Long flight endurance High-speed operation over long distances Cost-effective large-scale monitoring	Requires takeoff/landing space Less stable for low-altitude Harder to retrieve in case of failure Require precise sensors to deal with motion
	Hybrid VTOL		[76]	Combines vertical and horizontal flight Provides coverage over large areas Capable of vertical takeoff and landing More flexible deployment options	More complex system than multi-rotors Higher cost compared to quadcopters Requires specialized training and operation Increased maintenance demands